

Positive $R(\mathcal{V})$ Does Not Force Positive $d_{\mathcal{V}}$

Counterexample to Remark 2.23 on Fréchet approximation schemes

Run `fa_banach_001`, lane 18

11 August 2026

Status. Candidate full counterexample, likely valid, for human review. The proof is self-contained and analytic; novelty remains subject to broader literature review.

1 Source question

Let $(X, \|\cdot\|)$ be an infinite-dimensional Fréchet space and let $\mathcal{V} = (V_n)_{n \geq 1}$ be nested linear subspaces such that $\overline{V_n} \subsetneq V_{n+1}$ and $X = \overline{\bigcup_{n \geq 1} V_n}$. Define

$$\rho_n(x) = \text{dist}(x, V_n), \quad d_{n,\mathcal{V}} = \sup_{v \in V_{n+1}} \rho_n(v), \quad d_{\mathcal{V}} = \inf_{n \geq 1} d_{n,\mathcal{V}},$$

and

$$R(\mathcal{V}) = \inf_{v \in (\bigcup_n V_n) \setminus \{0\}} \sup_{t \in \mathbb{R}_+} \|tv\|.$$

Aksoy and Lewicki prove $d_{\mathcal{V}} > 0 \Rightarrow R(\mathcal{V}) > 0$. They note that the converse holds when every V_n is finite-dimensional and ask whether the equivalence holds for arbitrary $\mathcal{V}$ [1]. The answer is no.

Notice that the assumption 2.14 in Theorem 2.19 is necessary.

Example 2.21. Let X be a Banach space and let $W_n = \{x \in X : \|x\| \leq n\}$. Let $\{e_n\}$ be a decreasing sequence of positive numbers tending to zero. Since $X = \bigcup_{n=1}^{\infty} W_n$, there is no $x \in X$ such that

$$\frac{e_n}{3} \leq \text{dist}(x, W_n) \leq 3e_{n-1}.$$

At the end of this section we show some results concerning Fréchet spaces $(X, \|\cdot\|)$ and sequences of subspaces of X $\mathcal{V} = \{V_n\}$ satisfying or not satisfying the assumption 2.12. First notice that by [12], Prop. 3.4 and Cor. 3.8, if for all n , V_n are finite-dimensional, and $X = \overline{\bigcup_{n=1}^{\infty} V_n}$ then $d_{\mathcal{V}} > 0$ provided $R(\mathcal{V}) > 0$, where

$$R(\mathcal{V}) = \inf \left\{ \sup \{ \|tv\| : t \in \mathbb{R}_+, v \in \left(\bigcup_{n=1}^{\infty} V_n \right) \setminus \{0\} \} \right\}.$$

In general, we have the following:

Lemma 2.22. *Let $(X, \|\cdot\|)$ be an infinite-dimensional Fréchet space. Let $\mathcal{V} = \{V_n\}$ be a nested sequence of linear subspaces of X satisfying $\overline{V_n} \subseteq V_{n+1}$ such that*

$$X = \overline{\bigcup_{n=1}^{\infty} V_n}.$$

If $d_{\mathcal{V}} > 0$ then $R(\mathcal{V}) > 0$.

Proof. Notice that for any $x \in X$, $n \in \mathbb{N}$ and $t \in \mathbb{R}_+$, $\|tx\| \geq \rho_n(tx)$. Since $X = \overline{\bigcup_{n=1}^{\infty} V_n}$, this implies that

$$R(\mathcal{V}) \geq d_{\mathcal{V}},$$

which proves our claim. $\square$

Remark 2.23. *By [12], Prop. 3.4 and Cor. 3.8 and Lemma 2.22 if all n , V_n are finite-dimensional, then $d_{\mathcal{V}} > 0$ if and only if $R(\mathcal{V}) > 0$. We do not know if this is satisfied for arbitrary $\mathcal{V}$.*

Lemma 2.24. *Let $(X, \|\cdot\|)$ be an infinite-dimensional Fréchet space. Let $\mathcal{V} = \{V_n\}$ be a nested sequence of linear subspaces of X satisfying $\overline{V_n} \subseteq V_{n+1}$ such that*

$$X = \overline{\bigcup_{n=1}^{\infty} V_n}.$$

Let $\|\cdot\|_1$ be an F -norm defined on X equivalent to $\|\cdot\|$. Denote

Lemma 2.22 and Remark 2.23, source PDF page 16.

2 Idea of the proof

Use the usual smooth-function space, but metrize it by a bounded product F -norm. Along every nonzero ray, every coordinate of this metric saturates, so the supremum of the norm is exactly one; this forces $R(\mathcal{V}) = 1$. For the subspaces, filter functions by how many derivatives of their germ at zero may be nonzero. Passing from V_n to V_{n+1} adds just the n th jet. A degree n monomial times a shrinking cutoff removes that jet. Its low derivatives vanish in the limit, while all uncontrolled high derivatives live in a metric tail of total weight 2^{-n} . Thus $d_{n,\mathcal{V}} \leq 2^{-n}$.

3 The counterexample

Let $X = C^\infty([0, 1]; \mathbb{K})$, where $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, and set

$$p_j(f) = \max_{0 \leq k \leq j} \|f^{(k)}\|_\infty, \quad \|f\|_F = \sum_{j=0}^{\infty} 2^{-(j+1)} \frac{p_j(f)}{1 + p_j(f)}. \quad (1)$$

For $n \geq 1$, define

$$V_n = \{f \in X : f^{(k)}(0) = 0 \text{ for every } k \geq n\}. \quad (2)$$

Theorem 1. *The F-norm in (1) makes X a Fréchet space, and the subspaces in (2) are closed, strictly nested, and infinite-dimensional, with dense union. Nevertheless,*

$$R(\mathcal{V}) = 1 \quad \text{and} \quad d_{\mathcal{V}} = 0.$$

In particular, $R(\mathcal{V}) > 0$ does not imply $d_{\mathcal{V}} > 0$ for arbitrary $\mathcal{V}$.

4 Topology and density

The functions $s \mapsto s/(1+s)$ are increasing and subadditive on $[0, \infty)$, so (1) is an F-norm. It induces the usual C^∞ topology: convergence in the F-norm forces convergence in every p_j by looking at the j th summand, while convergence in every p_j implies convergence of the sum by splitting it into a finite head and a uniformly small geometric tail. Hence completeness is the usual completeness of $C^\infty[0, 1]$.

Every jet functional $f \mapsto f^{(k)}(0)$ is continuous. Thus V_n , an intersection of their kernels, is closed. We have $V_n \subset V_{n+1}$, and the inclusion is proper because $x \mapsto x^n$ lies in $V_{n+1} \setminus V_n$. Each V_n is infinite-dimensional because it contains every smooth function supported in a compact subset of $(0, 1]$.

The union contains every polynomial: a polynomial of degree d belongs to V_{d+1} . For completeness, we verify that polynomials are dense in the full smooth topology. Fix $f \in X$, $m \geq 1$, and $\varepsilon > 0$. By the Weierstrass theorem choose a polynomial q with $\|f^{(m)} - q\|_\infty < \varepsilon$. Define the polynomial

$$P(x) = \sum_{r=0}^{m-1} \frac{f^{(r)}(0)}{r!} x^r + \int_0^x \frac{(x-s)^{m-1}}{(m-1)!} q(s) ds.$$

Then $P^{(m)} = q$ and $P^{(r)}(0) = f^{(r)}(0)$ for $r < m$. Repeated integration gives, for $0 \leq r < m$,

$$\|f^{(r)} - P^{(r)}\|_\infty \leq \frac{\|f^{(m)} - q\|_\infty}{(m-r)!} < \varepsilon.$$

Thus $p_m(f - P) < \varepsilon$. The case $m = 0$ is Weierstrass approximation itself. Hence $X = \overline{\bigcup_n V_n}$.

5 Exact value of $R(\mathcal{V})$

If $f \neq 0$, then $p_j(f) \geq p_0(f) > 0$ for all j . For $t \geq 0$,

$$\|tf\|_F = \sum_{j=0}^{\infty} 2^{-(j+1)} \frac{tp_j(f)}{1 + tp_j(f)} \leq 1.$$

Every summand increases to its full weight as $t \rightarrow \infty$. Monotone convergence therefore yields

$$\lim_{t \rightarrow \infty} \|tf\|_F = \sum_{j=0}^{\infty} 2^{-(j+1)} = 1.$$

Consequently $\sup_{t \in \mathbb{R}_+} \|tf\|_F = 1$ for every nonzero f , in particular for every nonzero $f \in \bigcup_n V_n$. Hence

$$R(\mathcal{V}) = 1. \quad (3)$$

6 Decay of $d_{n,\mathcal{V}}$

Fix $n \geq 1$ and $f \in V_{n+1}$. Put $a = f^{(n)}(0)$. Choose $\chi \in C^\infty([0, \infty))$ such that $\chi = 1$ near zero and $\chi = 0$ on $[1, \infty)$. For $0 < \delta \leq 1$, define

$$h_\delta(x) = a \frac{x^n}{n!} \chi(x/\delta). \quad (4)$$

Near zero, $h_\delta(x) = ax^n/n!$. Therefore

$$h_\delta^{(n)}(0) = a, \text{quad} h_\delta^{(k)}(0) = 0 \quad (k > n).$$

Because $f \in V_{n+1}$ already has $f^{(k)}(0) = 0$ for $k > n$, it follows that $g_\delta = f - h_\delta \in V_n$.

For every fixed $k < n$, Leibniz' rule and the support condition give a constant $C_{n,k,\chi}$, independent of a and δ , such that

$$\|h_\delta^{(k)}\|_\infty \leq C_{n,k,\chi} |a| \delta^{n-k}. \quad (5)$$

Indeed, each term is a constant multiple of $x^{n-r} \delta^{-(k-r)} \chi^{(k-r)}(x/\delta)$ on $0 \leq x \leq \delta$, and hence has size $O(\delta^{n-k})$. It follows that $p_j(h_\delta) \rightarrow 0$ for every $j < n$.

Using $s/(1+s) \leq 1$ on the remaining terms, we obtain

$$\begin{aligned} \rho_n(f) &\leq \|f - g_\delta\|_F = \|h_\delta\|_F, \\ \limsup_{\delta \downarrow 0} \|h_\delta\|_F &\leq \sum_{j=n}^{\infty} 2^{-(j+1)} = 2^{-n}. \end{aligned}$$

Thus $\rho_n(f) \leq 2^{-n}$ for every $f \in V_{n+1}$. Taking the supremum,

$$d_{n,\mathcal{V}} \leq 2^{-n};$$

and therefore

$$d_{\mathcal{V}} = \inf_{n \geq 1} d_{n,\mathcal{V}} = 0. \quad (6)$$

Equations (3) and (6) prove Theorem 1.

7 Verification, scope, and novelty status

The proof uses no computation. The only limiting estimate relevant to the supremum in $d_{n,\mathcal{V}}$ is uniform in the required direction: for every fixed $f \in V_{n+1}$ it proves the same upper bound 2^{-n} , even though the cutoff scale may depend on $f^{(n)}(0)$. If that jet is zero, then $f \in V_n$ already.

The construction works over both real and complex scalars. All V_n are closed and infinite-dimensional, so the failure is precisely outside the finite-dimensional hypothesis highlighted in Remark 2.23. It does not contradict the proved implication $d_{\mathcal{V}} > 0 \Rightarrow R(\mathcal{V}) > 0$.

A bounded search of the run indexes and exact web queries found the source paper and its journal version, but no later resolution of Remark 2.23. Priority is not asserted. Expert proof review and a broader citation search are recommended before dissemination.

References

- [1] A. G. Aksoy and G. Lewicki, *Bernstein's Lethargy Theorem in Fréchet Spaces*, arXiv:1503.06190 (2015).